

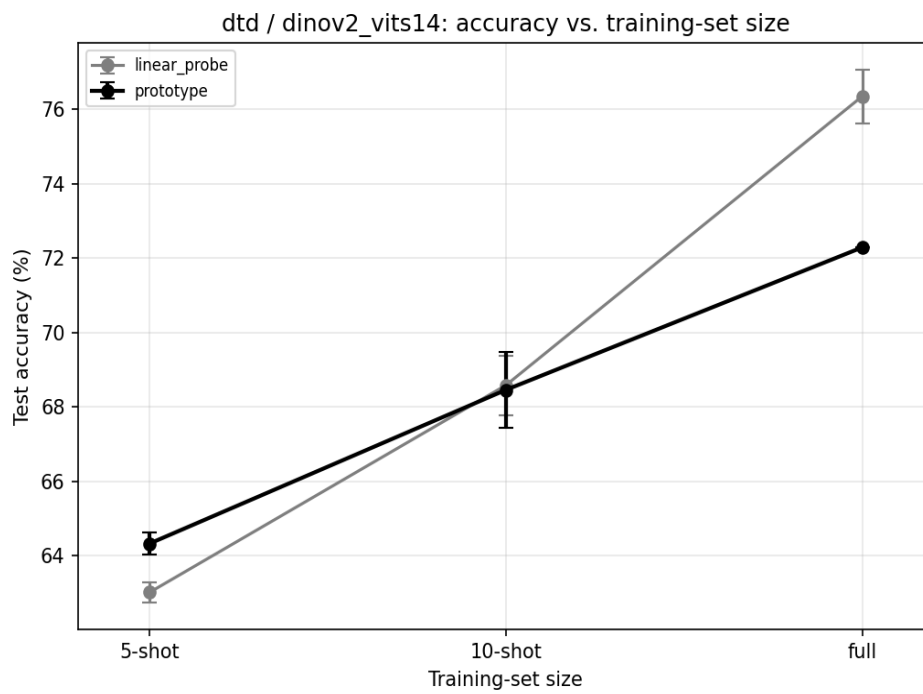
Stage 1 and Stage 2 Results

Dataset	Encoder	Method	T	K-shot	Runs	Test Accuracy	Delta
dtd	dinov2_vits14	linear_probe	-	5	3	63.01% +/- 0.27%	-
dtd	dinov2_vits14	linear_probe	-	10	3	68.58% +/- 0.80%	-
dtd	dinov2_vits14	linear_probe	-	full	3	76.35% +/- 0.72%	-
dtd	dinov2_vits14	prototype	-	5	3	64.33% +/- 0.30%	-
dtd	dinov2_vits14	prototype	-	10	3	68.46% +/- 1.01%	-
dtd	dinov2_vits14	prototype	-	full	1	72.29%	-
dtd	dinov2_vits14	fm_standard	4	5	3	61.31% +/- 0.75%	-3.01% +/- 0.46%
dtd	dinov2_vits14	fm_standard	12	5	3	61.70% +/- 0.83%	-2.62% +/- 0.55%
dtd	dinov2_vits14	fm_standard	4	10	3	65.60% +/- 0.59%	-2.85% +/- 1.45%
dtd	dinov2_vits14	fm_standard	12	10	3	65.92% +/- 0.57%	-2.54% +/- 1.48%
dtd	dinov2_vits14	fm_standard	4	full	1	74.63%	+2.34%
dtd	dinov2_vits14	fm_standard	12	full	1	74.79%	+2.50%
dtd	dinov2_vits14	fm_rolled	4	5	3	57.48% +/- 1.32%	-6.84% +/- 1.02%
dtd	dinov2_vits14	fm_rolled	12	5	3	57.50% +/- 1.04%	-6.83% +/- 0.73%
dtd	dinov2_vits14	fm_rolled	4	10	3	61.74% +/- 0.48%	-6.72% +/- 1.46%
dtd	dinov2_vits14	fm_rolled	12	10	3	61.74% +/- 0.27%	-6.72% +/- 1.28%
dtd	dinov2_vits14	fm_rolled	4	full	1	73.51%	+1.22%
dtd	dinov2_vits14	fm_rolled	12	full	1	72.93%	+0.64%
dtd	resnet18	linear_probe	-	5	3	45.80% +/- 0.75%	-
dtd	resnet18	linear_probe	-	10	3	52.39% +/- 0.74%	-
dtd	resnet18	linear_probe	-	full	3	62.55% +/- 0.51%	-
dtd	resnet18	prototype	-	5	3	46.93% +/- 0.32%	-
dtd	resnet18	prototype	-	10	3	51.97% +/- 0.54%	-
dtd	resnet18	prototype	-	full	1	58.78%	-
dtd	resnet18	fm_standard	4	5	3	44.82% +/- 1.45%	-2.11% +/- 1.19%
dtd	resnet18	fm_standard	12	5	3	45.14% +/- 1.36%	-1.79% +/- 1.07%
dtd	resnet18	fm_standard	4	10	3	48.65% +/- 1.02%	-3.32% +/- 0.52%
dtd	resnet18	fm_standard	12	10	3	49.13% +/- 1.17%	-2.84% +/- 0.65%
dtd	resnet18	fm_standard	4	full	1	59.52%	+0.74%
dtd	resnet18	fm_standard	12	full	1	59.57%	+0.80%
dtd	resnet18	fm_rolled	4	5	3	40.51% +/- 0.75%	-6.42% +/- 0.45%
dtd	resnet18	fm_rolled	12	5	3	40.60% +/- 0.85%	-6.33% +/- 0.53%

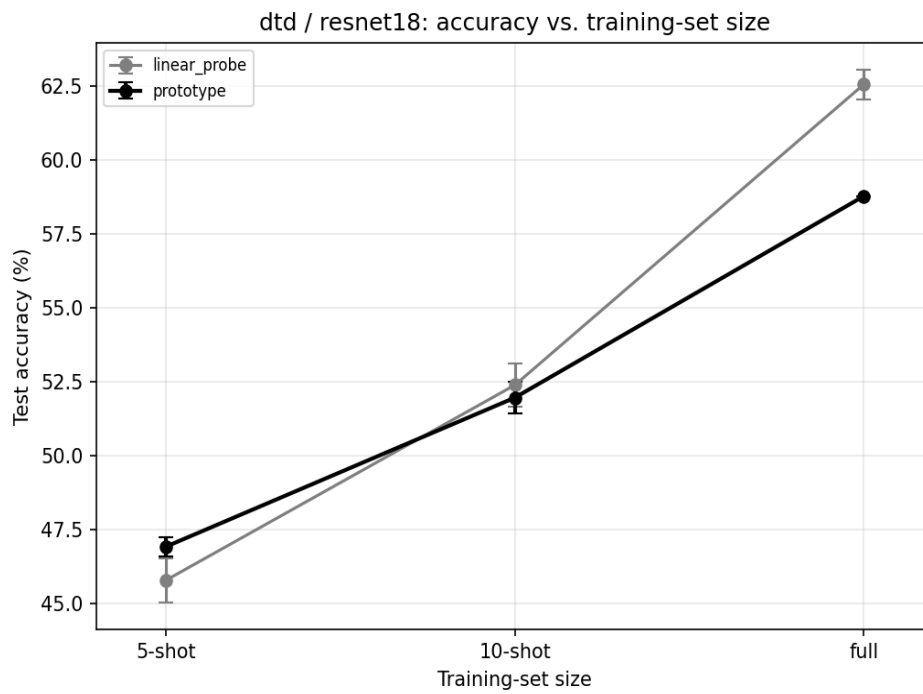
Dataset	Encoder	Method	T	K-shot	Runs	Test Accuracy	Delta
dtd	resnet18	fm_rolled	4	10	3	40.62% +/- 1.00%	-11.35% +/- 1.20%
dtd	resnet18	fm_rolled	12	10	3	41.65% +/- 0.90%	-10.32% +/- 0.88%
dtd	resnet18	fm_rolled	4	full	1	56.06%	-2.71%
dtd	resnet18	fm_rolled	12	full	1	56.01%	-2.77%
flowers102	resnet18	linear_probe	-	5	3	75.13% +/- 0.22%	-
flowers102	resnet18	linear_probe	-	10	3	83.22% +/- 0.14%	-
flowers102	resnet18	linear_probe	-	full	3	83.22% +/- 0.14%	-
flowers102	resnet18	prototype	-	5	3	69.62% +/- 0.32%	-
flowers102	resnet18	prototype	-	10	3	75.22% +/- 0.00%	-
flowers102	resnet18	prototype	-	full	1	75.22%	-
flowers102	resnet18	fm_standard	4	5	3	69.19% +/- 1.02%	-0.43% +/- 0.85%
flowers102	resnet18	fm_standard	12	5	3	69.59% +/- 1.01%	-0.02% +/- 0.82%
flowers102	resnet18	fm_standard	4	10	3	76.83% +/- 0.11%	+1.62% +/- 0.11%
flowers102	resnet18	fm_standard	12	10	3	77.17% +/- 0.21%	+1.95% +/- 0.21%
flowers102	resnet18	fm_standard	4	full	1	76.79%	+1.58%
flowers102	resnet18	fm_standard	12	full	1	76.94%	+1.72%
flowers102	resnet18	fm_rolled	4	5	3	62.54% +/- 1.32%	-7.07% +/- 1.12%
flowers102	resnet18	fm_rolled	12	5	3	63.07% +/- 0.68%	-6.54% +/- 0.40%
flowers102	resnet18	fm_rolled	4	10	3	72.46% +/- 0.38%	-2.76% +/- 0.38%
flowers102	resnet18	fm_rolled	12	10	3	72.65% +/- 0.37%	-2.57% +/- 0.37%
flowers102	resnet18	fm_rolled	4	full	1	72.89%	-2.33%
flowers102	resnet18	fm_rolled	12	full	1	73.07%	-2.15%

Accuracy vs. training-set size

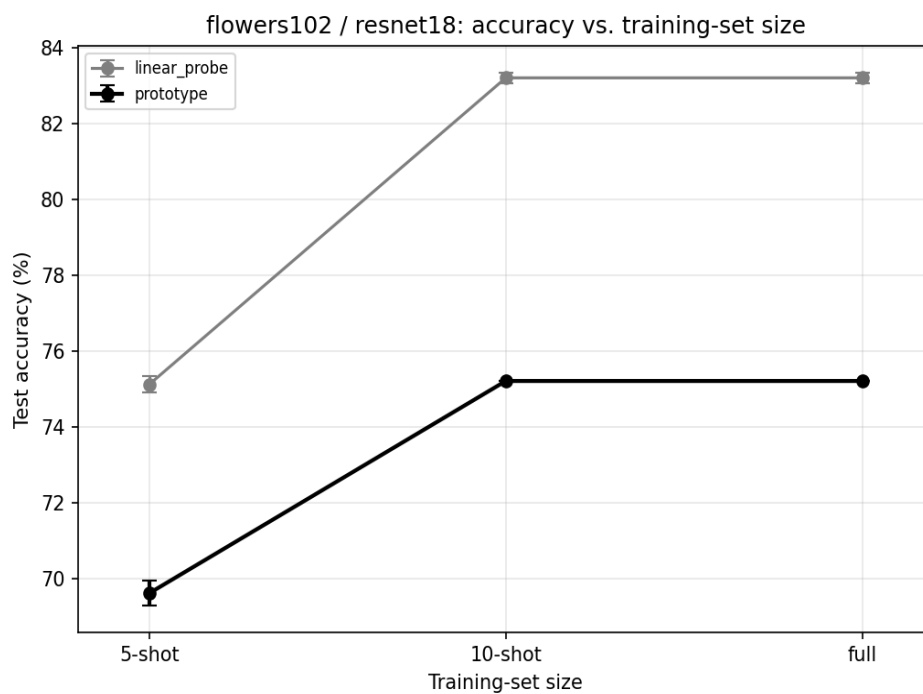
dtd / dinov2_vits14



dtd / resnet18

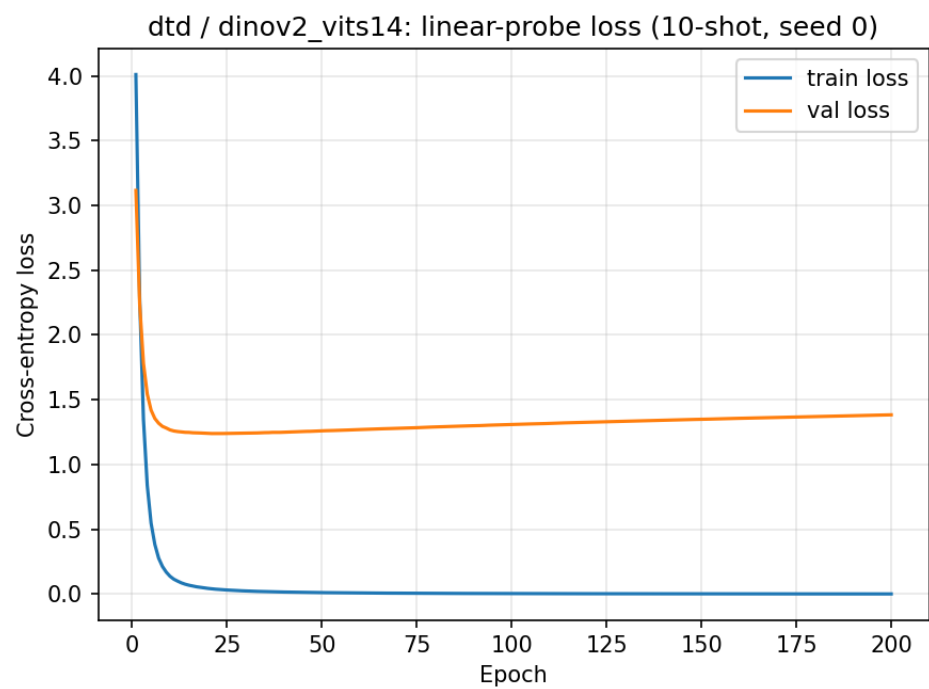


flowers102 / resnet18

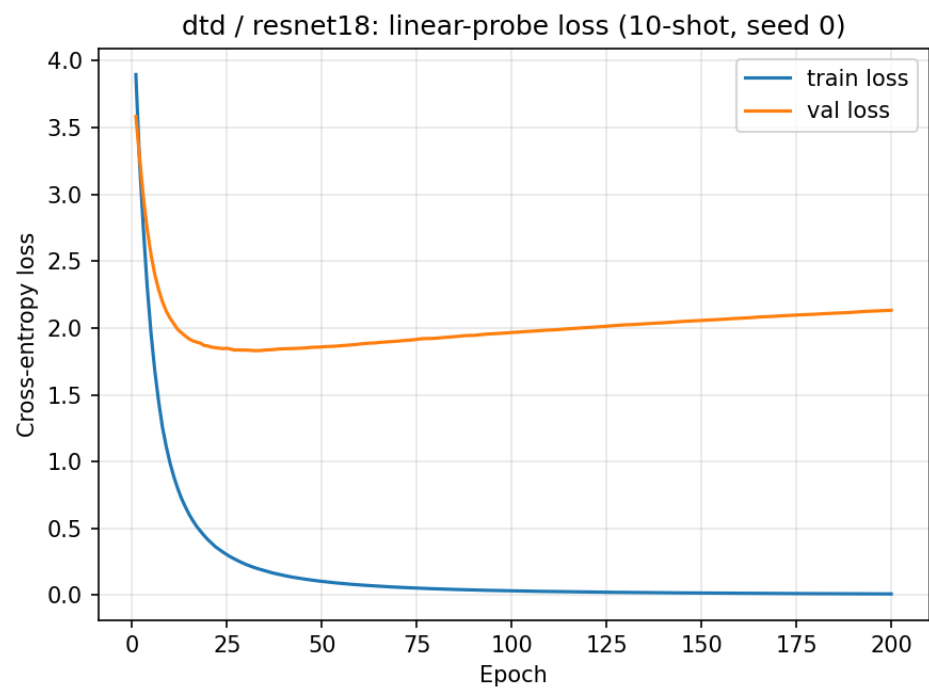


Training / validation loss (10-shot, seed 0)

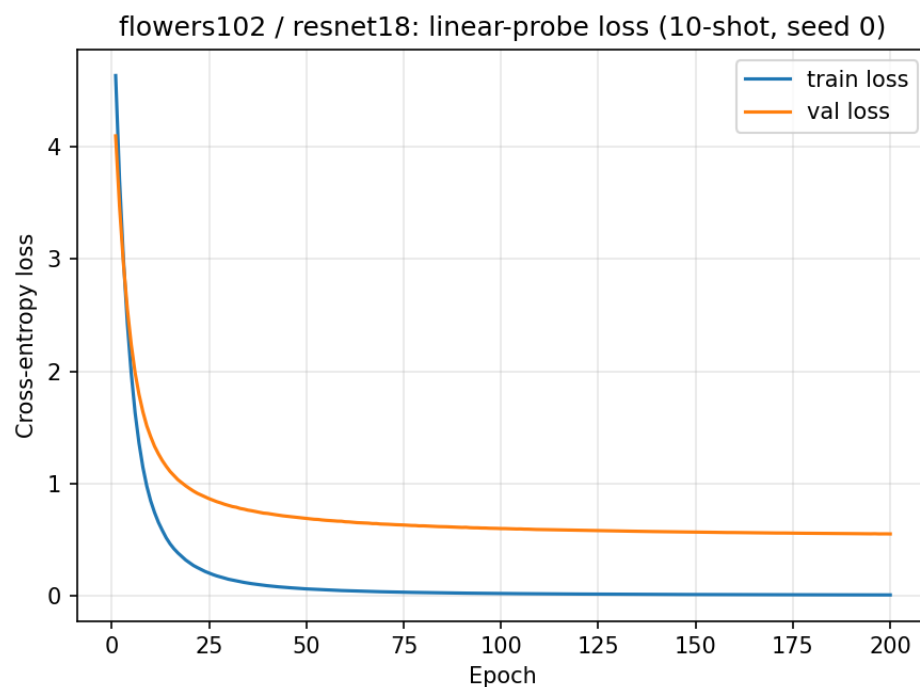
dtd / dinov2_vits14



dtd / resnet18



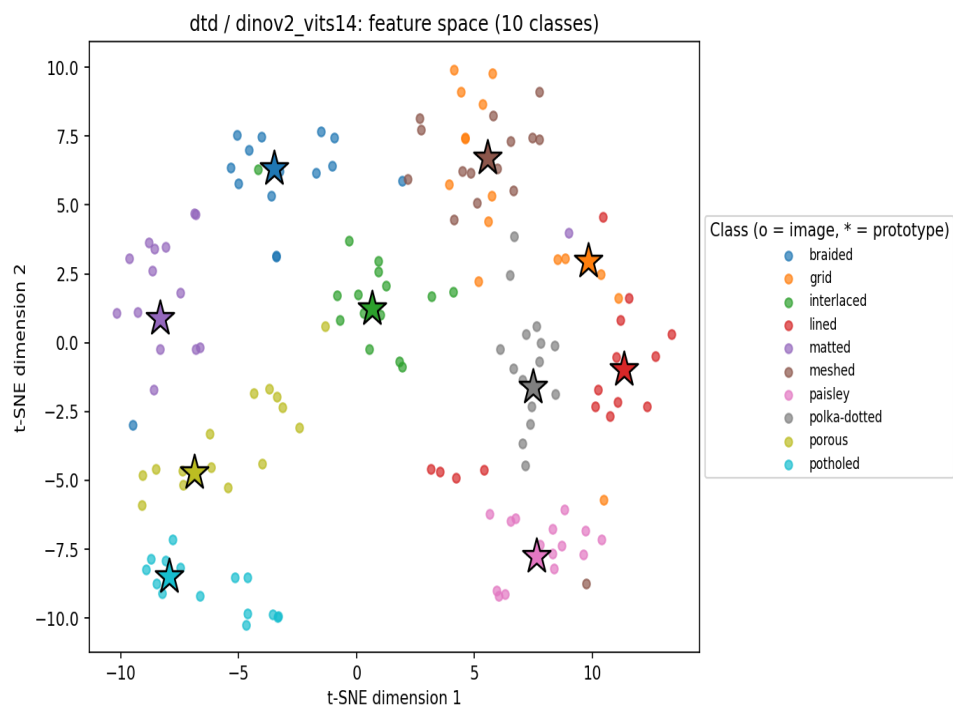
flowers102 / resnet18



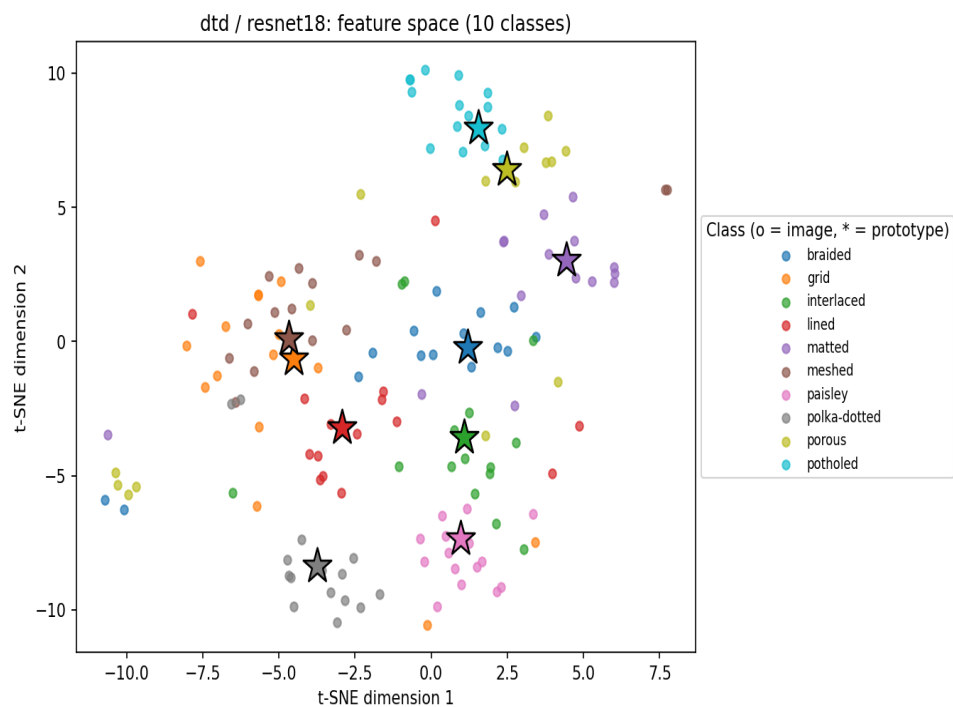
Confusion matrices (row-normalized)

dtd / dinov2_vits14

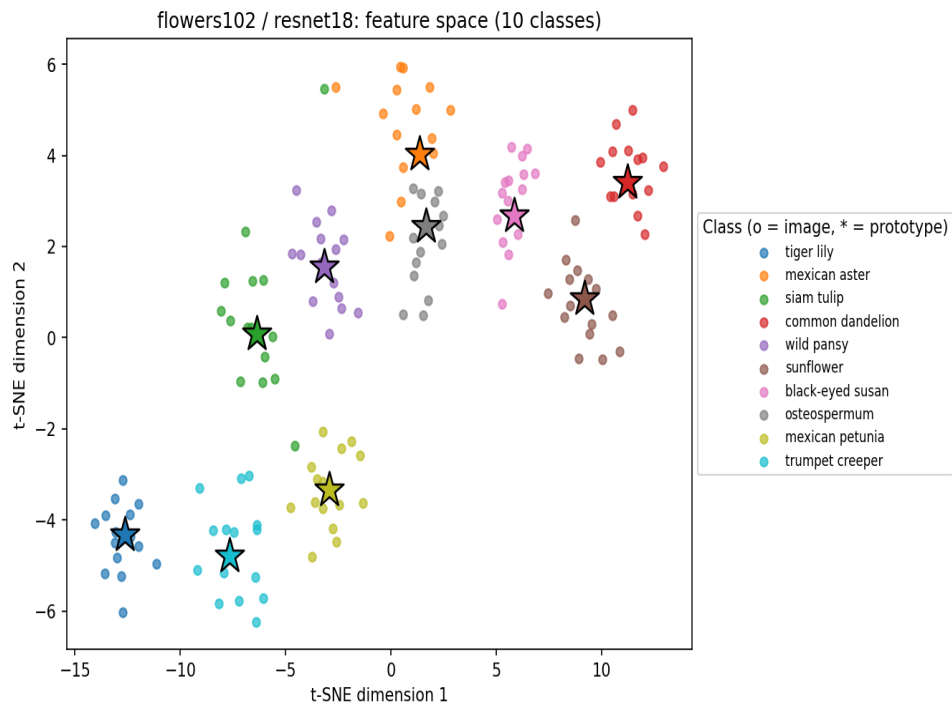
dtd / dinov2_vits14



dtd / resnet18

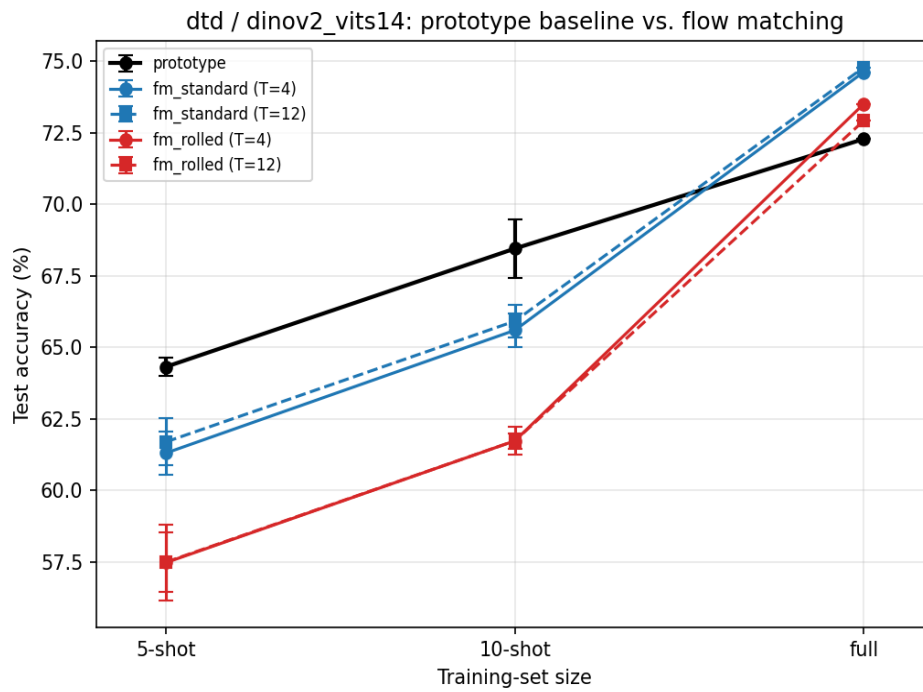


flowers102 / resnet18

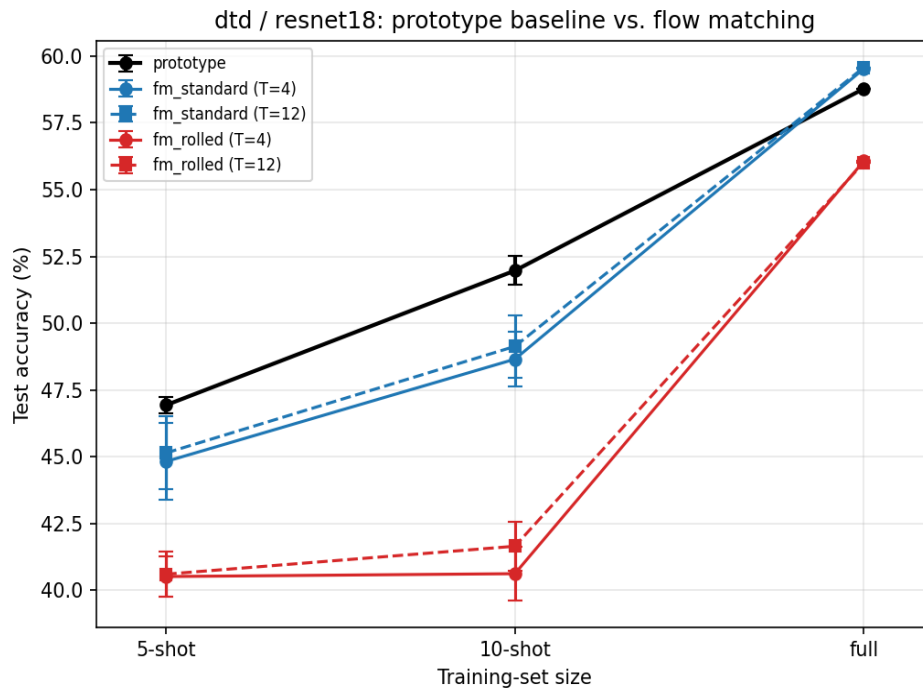


Stage 2: accuracy vs. training-set size

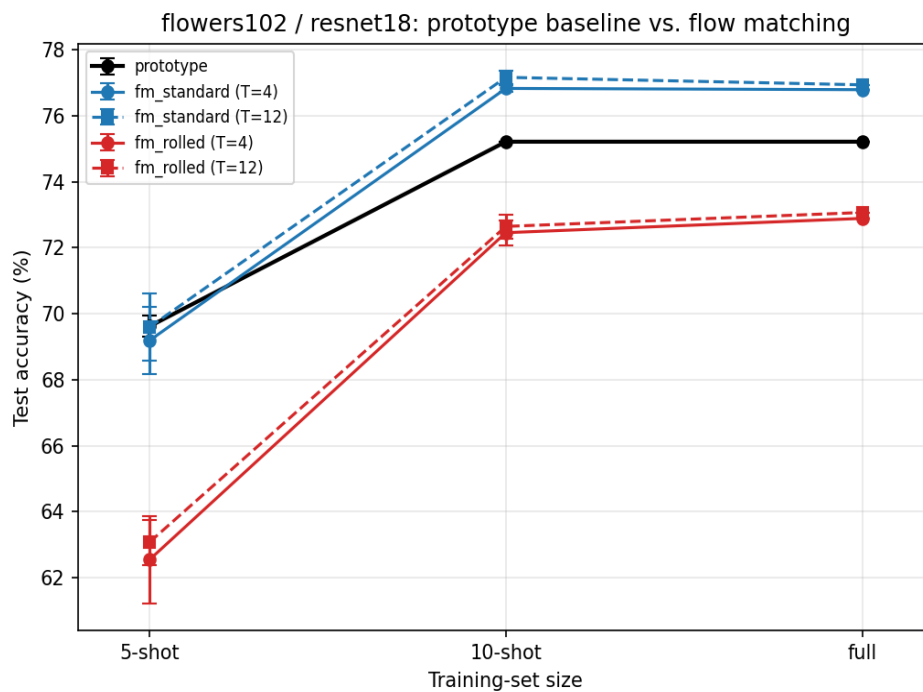
dtd / dinov2_vits14



dtd / resnet18

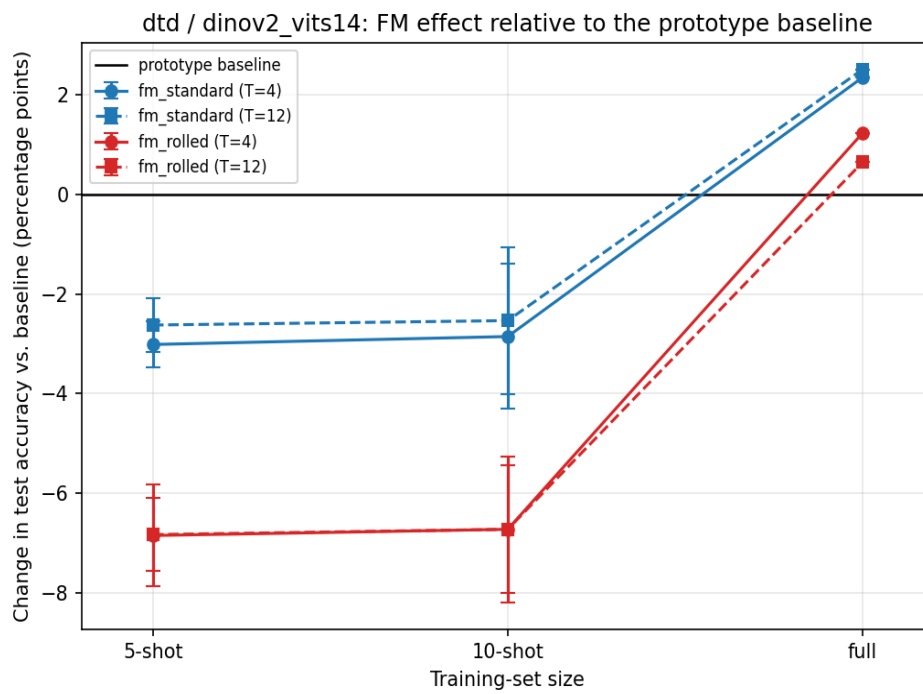


flowers102 / resnet18

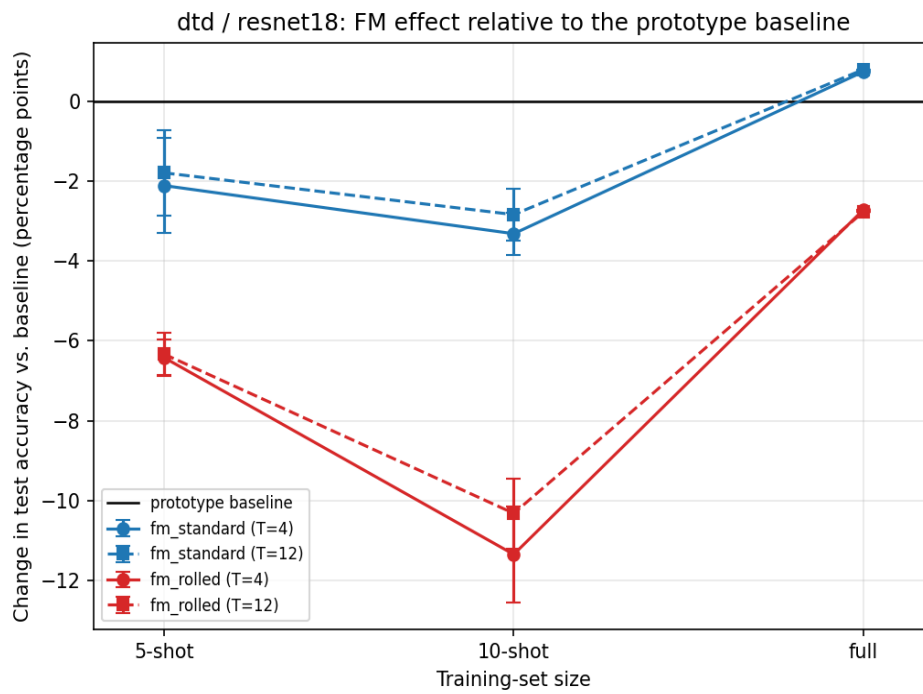


Stage 2: change in accuracy relative to the prototype baseline

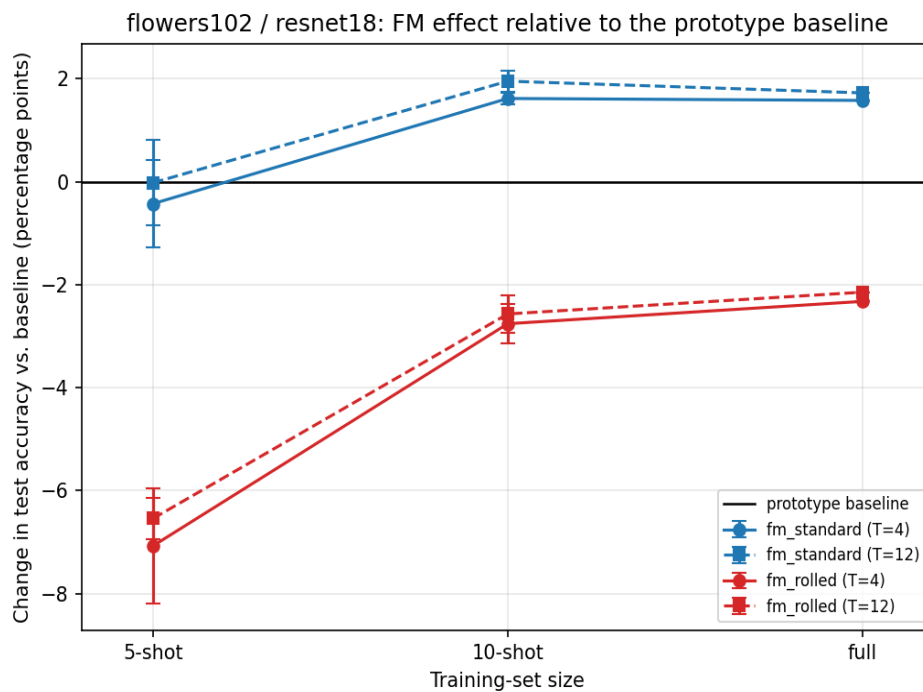
dtd / dinov2_vits14



dtd / resnet18



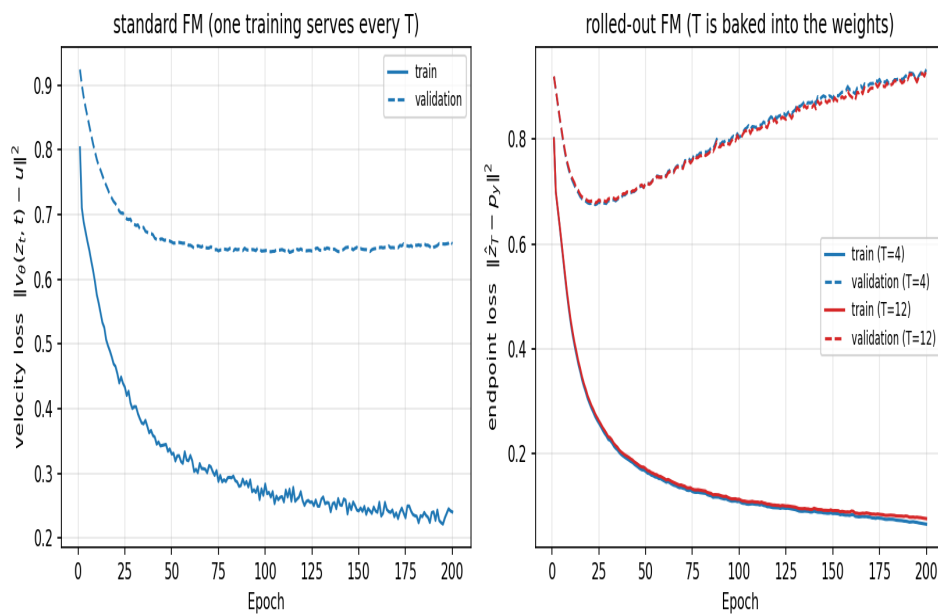
flowers102 / resnet18



Stage 2: flow-matching training curves (10-shot, seed 0)

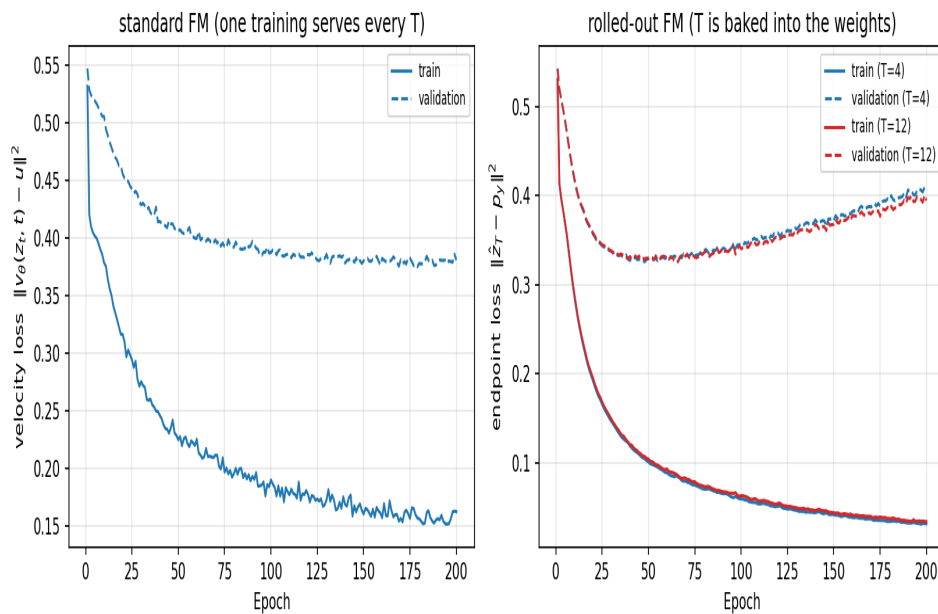
dtd / dinov2_vits14

dtd / dinov2_vits14: flow-matching training (10-shot, seed 0)
 note: the two panels measure different quantities and are not comparable to each other



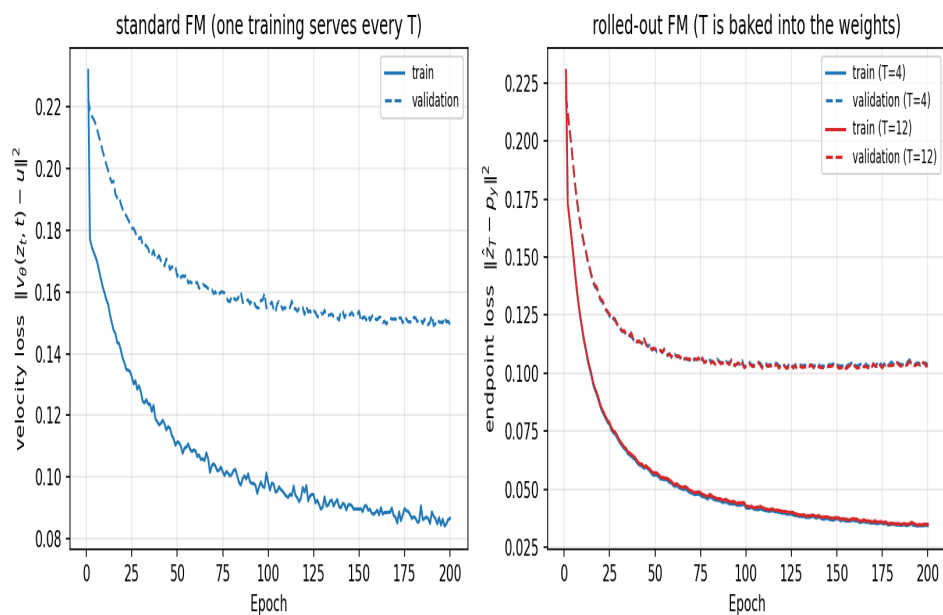
dtd / resnet18

dtd / resnet18: flow-matching training (10-shot, seed 0)
 note: the two panels measure different quantities and are not comparable to each other



flowers102 / resnet18

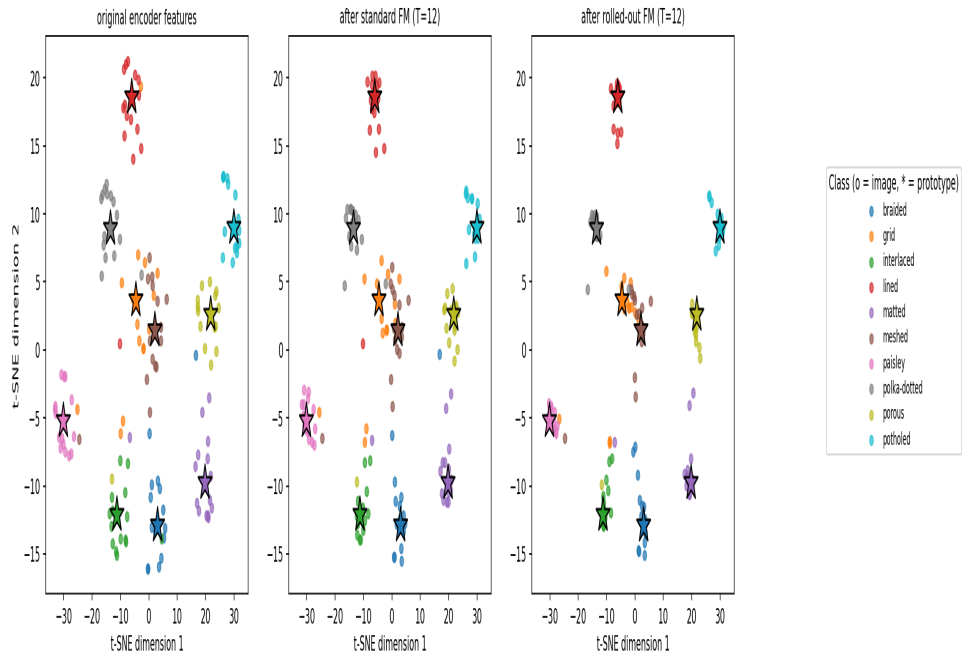
flowers102 / resnet18: flow-matching training (10-shot, seed 0)
note: the two panels measure different quantities and are not comparable to each other



Stage 2: feature space before and after flow matching (t-SNE)

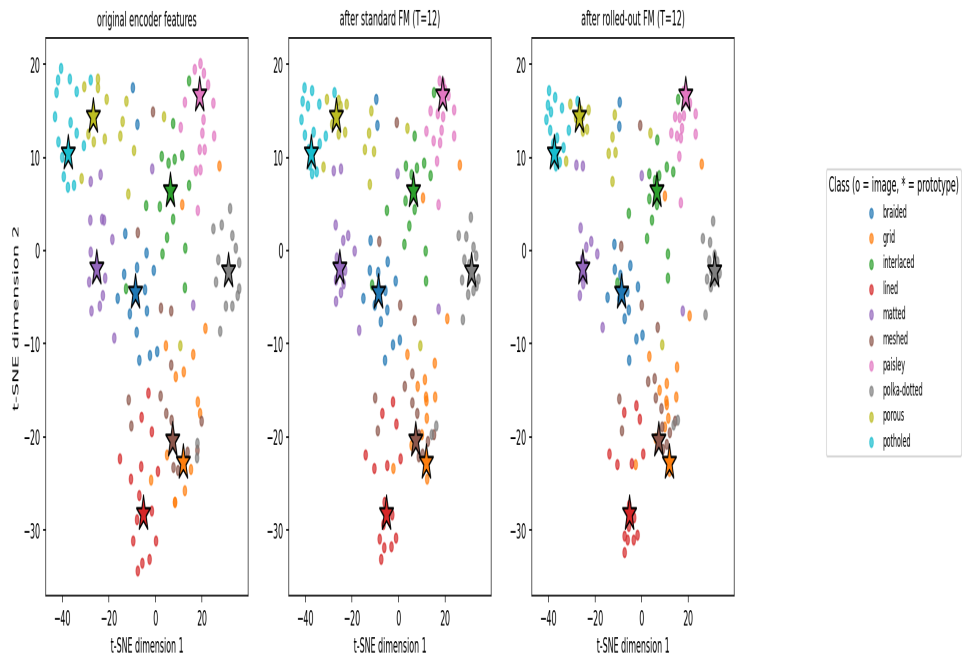
dtd / dinov2_vits14

dtd / dinov2_vits14: feature space before and after flow matching (K=full, seed 0, 10 classes)

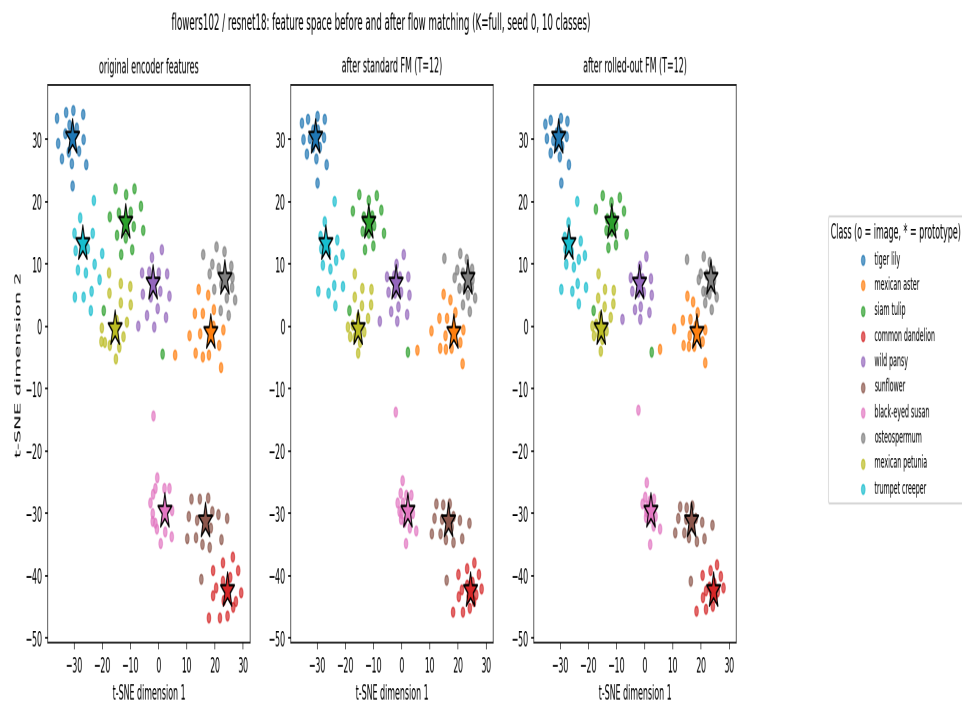


dtd / resnet18

dtd / resnet18: feature space before and after flow matching (K=full, seed 0, 10 classes)



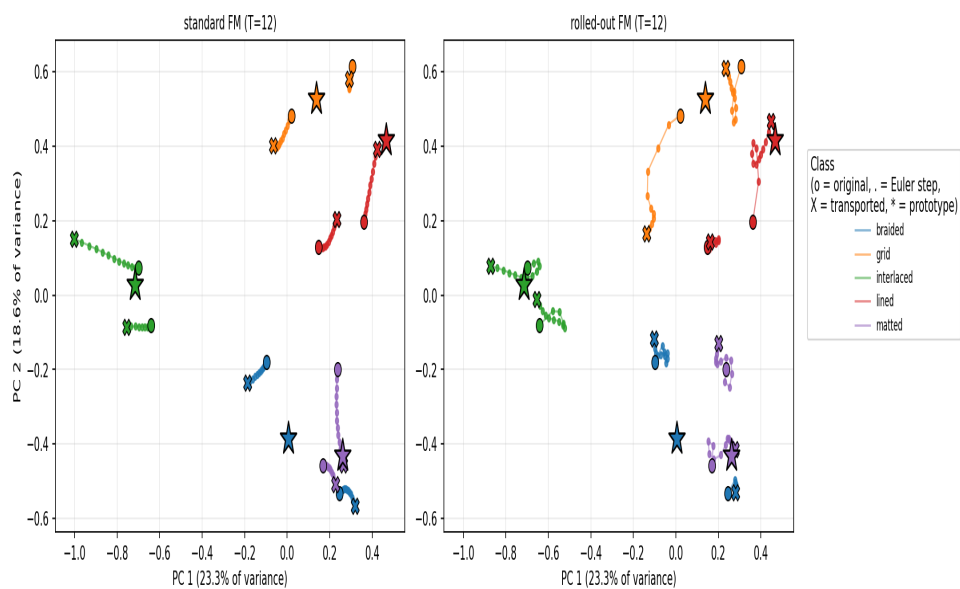
flowers102 / resnet18



Stage 2: flow trajectories (PCA)

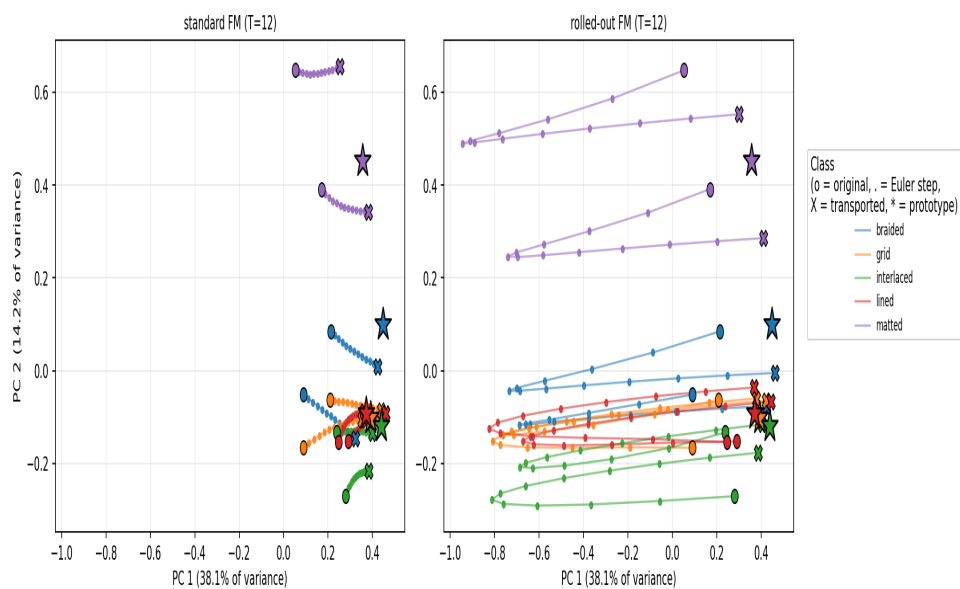
dtd / dinov2_vits14

dtd / dinov2_vits14: Euler flow trajectories (K=full, seed 0, 10 test examples)



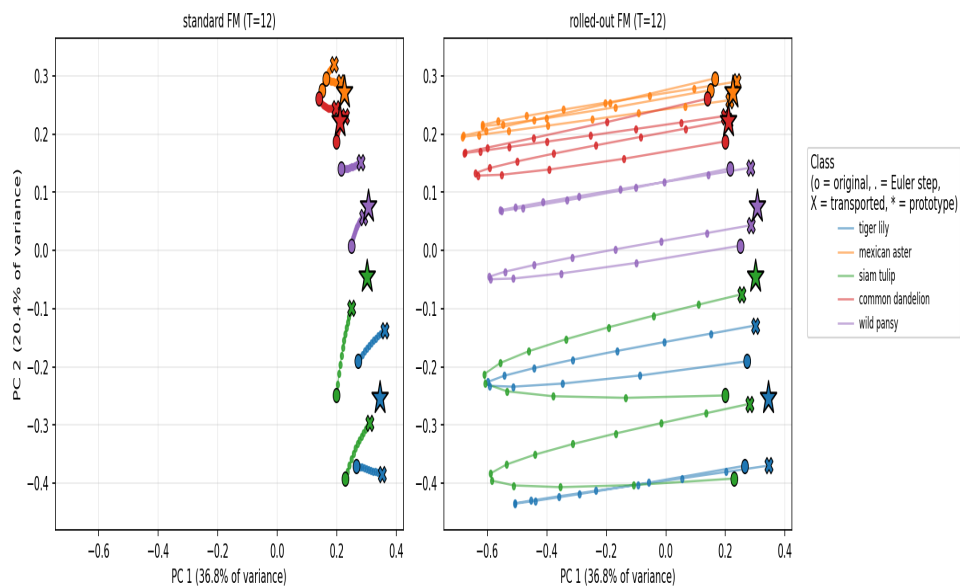
dtd / resnet18

dtd / resnet18: Euler flow trajectories (K=full, seed 0, 10 test examples)



flowers102 / resnet18

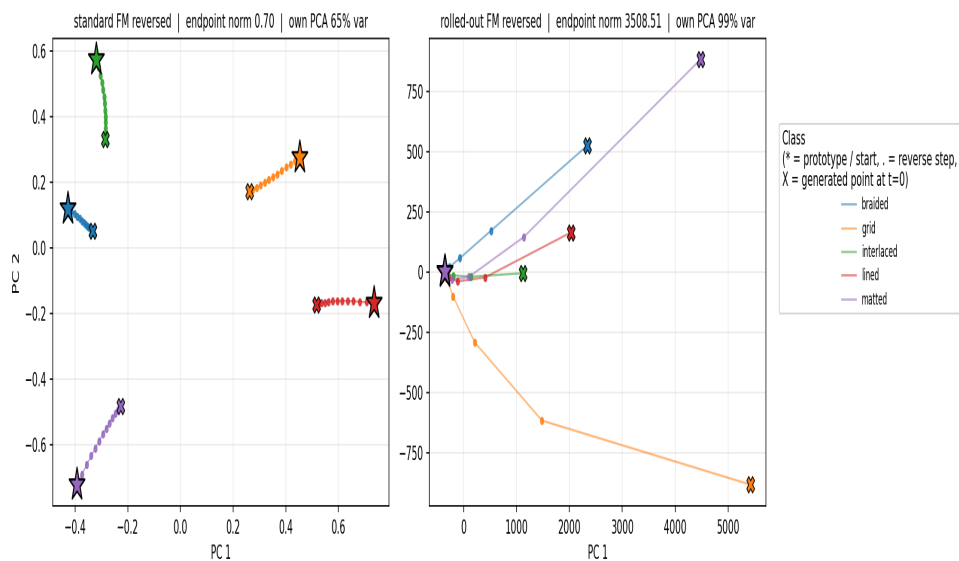
flowers102 / resnet18: Euler flow trajectories (K=full, seed 0, 10 test examples)



Stage 2: reverse flow from the class prototypes (PCA)

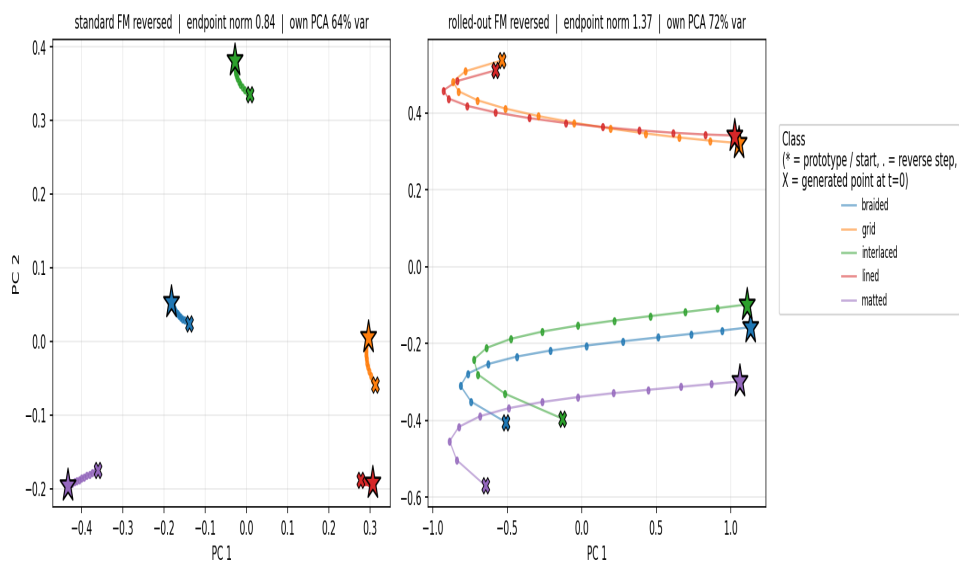
dtd / dinov2_vits14

dtd / dinov2_vits14: reverse flow from the class prototypes (K=full, seed 0, T=12)
 each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



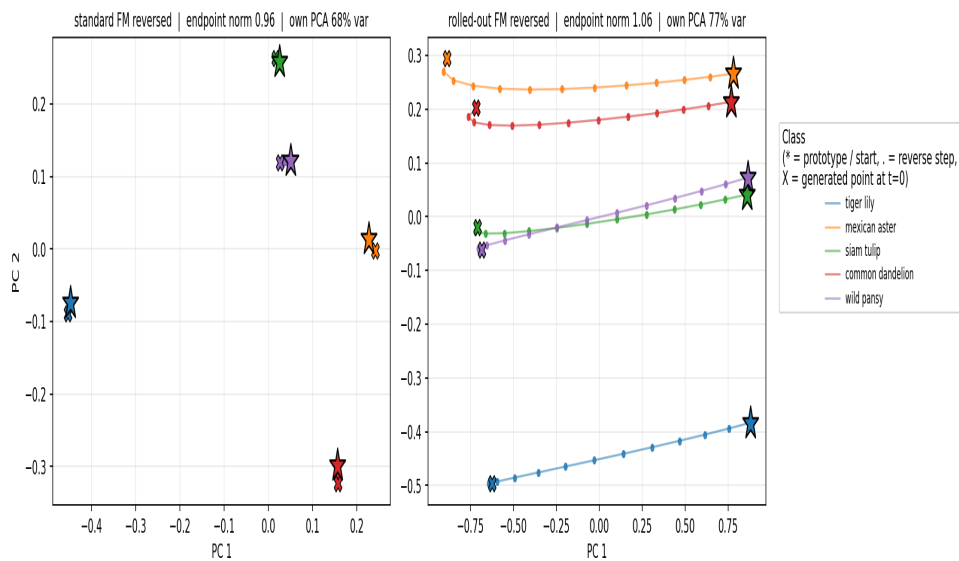
dtd / resnet18

dtd / resnet18: reverse flow from the class prototypes (K=full, seed 0, T=12)
 each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



flowers102 / resnet18

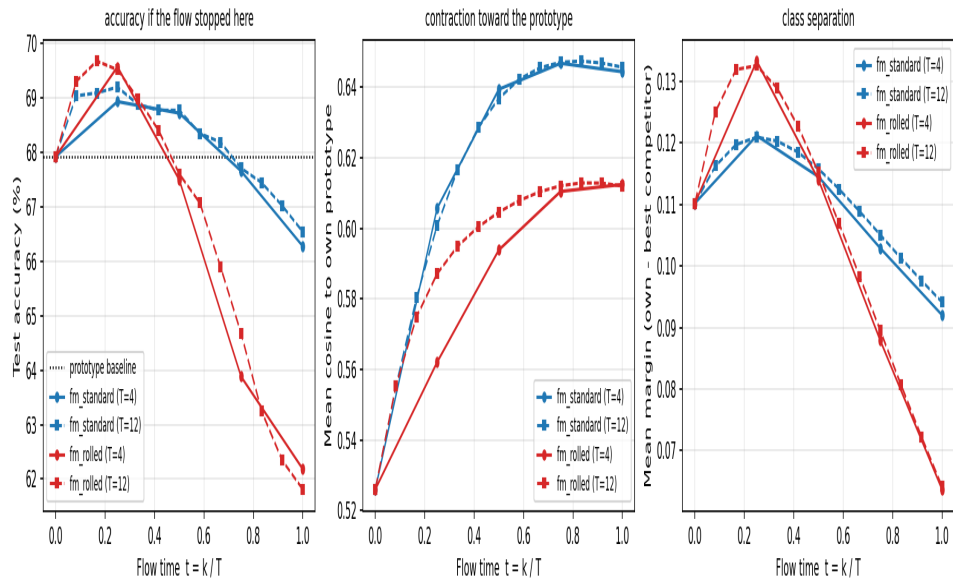
flowers102 / resnet18: reverse flow from the class prototypes (K=full, seed 0, T=12)
each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



Stage 2: metrics along the flow

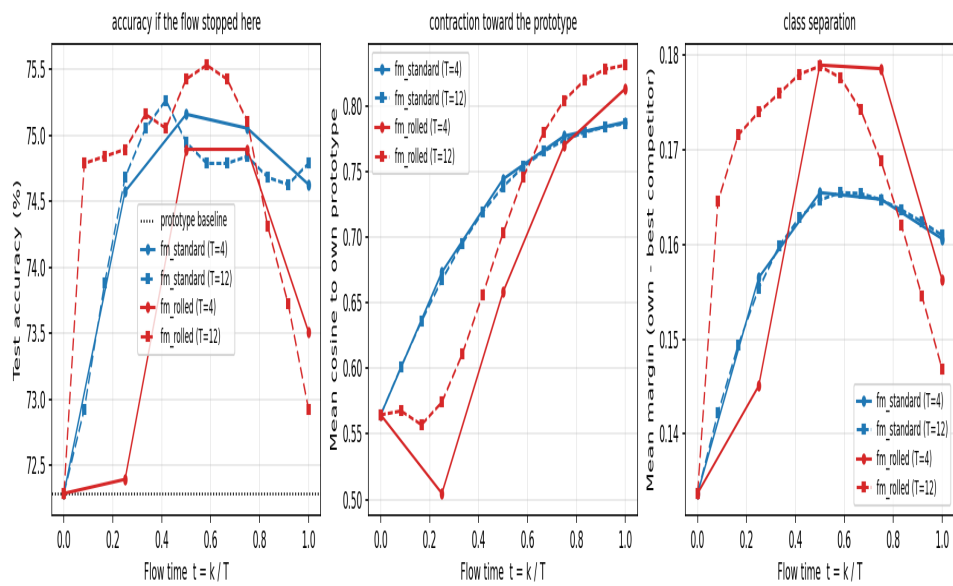
dtd (K=10) / dinov2_vits14

dtd / dinov2_vits14: metrics along the flow (K=10)



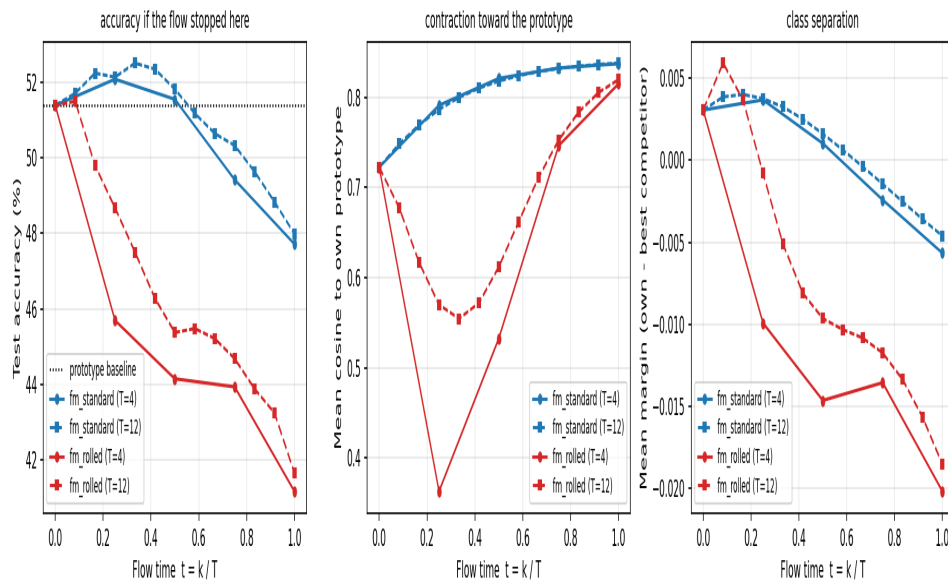
dtd (K=full) / dinov2_vits14

dtd / dinov2_vits14: metrics along the flow (K=full)



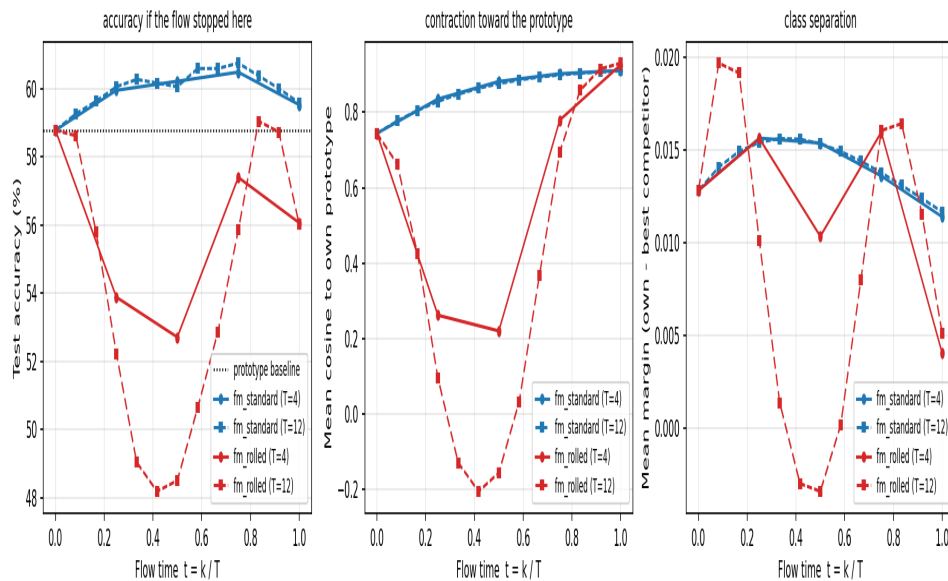
dtd (K=10) / resnet18

dtd / resnet18: metrics along the flow (K=10)



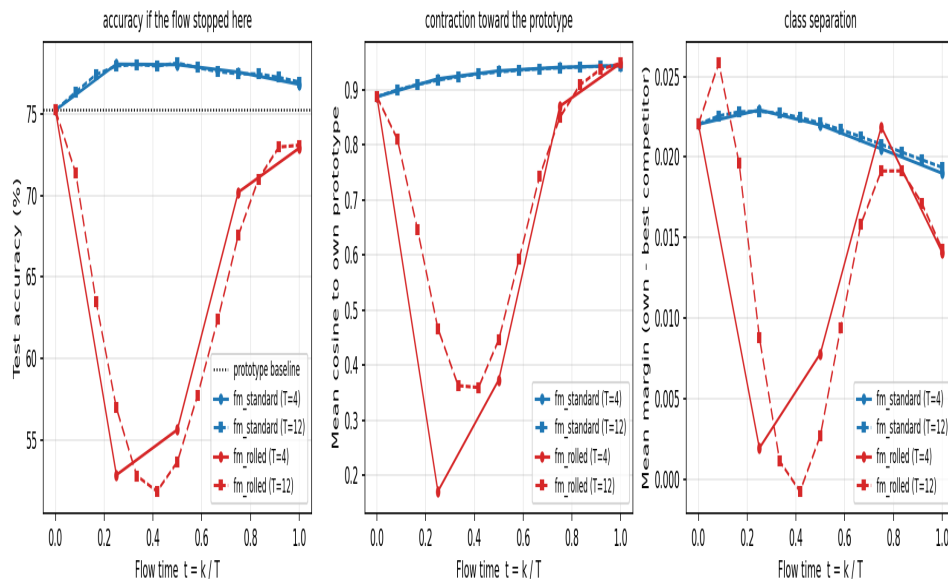
dtd (K=full) / resnet18

dtd / resnet18: metrics along the flow (K=full)



flowers102 (K=10) / resnet18

flowers102 / resnet18: metrics along the flow (K=10)



flowers102 (K=full) / resnet18

flowers102 / resnet18: metrics along the flow (K=full)

